

Lab 3: Confounding & Effect Measure Collapsibility

Overview

In this lab you will explore confounding and effect measure collapsibility using the Right Heart Catheterization (RHC) dataset, investigating the association between receiving a right-heart catheterization and all-cause mortality. You will prepare the data, produce a descriptive summary table, and fit a series of regression models to calculate and compare crude, conditional, and marginal effect estimates (odds ratios, risk ratios, and risk differences) — seeing how adjustment affects each measure differently. Your TA will walk through the worked solution during the session.

This is the workflow you will reuse in **P2** on your own locked paper.

Materials

- **Exercise instructions:** Confounding
- **Worked solution:** The R codes

Your task

1. Follow along as the TA explains the worked solution.
2. Run the R Markdown file yourself, as a group, on your own laptop.
3. Put your **group name and the first names of everyone present** in the `author:` field at the top of the file.
4. Add `sessionInfo()` in a final code chunk.
5. Knit to **PDF or HTML** and submit that one file before the lab ends.

No written explanations are required. If you get stuck, ask the TA in the room — that is what the session is for.

Deliverable

- **One file per group:** the knitted PDF or HTML.
- **Due:** end of the lab session (Thu Oct 1).

Grading: Complete / Incomplete

You receive **Complete** if the document knitted without errors, your group name and members are in it, and it was submitted before the lab ended. Only members present in the room receive credit.

Where this lab fits

This lab is one step in a weekly chain: Tuesday's lecture sets up the method, Thursday's lab runs it on fixed teaching data, and your increment applies it to the paper your group locked at M0.

	Document	How this lab connects
This week's lecture	Week 4 — Confounding and DAGs; Table 1; variable roles	Tuesday moved from a frozen cohort to estimates — what Table 1 is for, how to decide an adjustment set from causal and temporal reasoning, and why what you leave out (mediators, colliders) matters as much as what you put in. This lab runs that sequence and watches crude, conditional and marginal estimates behave differently.
Slide deck	Week 4 deck	The same material as slides, with speaker notes. Press S for notes, F for fullscreen.
Your increment	P2 — Confounding diagnostic	Produce your own Table 1 plus a written adjustment rationale — which variables you adjust for, which you deliberately do not, and why.
Worked example	P2 example — its code	A completed P2: the Table 1, and the adjustment rationale written out. Read the rationale first — it is the part that carries the marks.
Milestone	M1 — Replication and interrogation	P2 enters M1 as Confounding, adjustment set, Table 1 . P2 is optional and ungraded with nothing to submit; M1 is where it is assessed.